

Program : Diploma in Information Technology	
Course Code : 5261	Course Title: Network Routing Technologies
Semester : 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide basics of Network Routing Technologies from a Network Administrator's point of view.
- To provide the basics of a Network Router and its Administration.

Course Prerequisites:

Topic	Course Code	Course Title	Semester
Subnetting, WAN, IP Address, Routing		Fundamentals of Network Administration	3
IOS basics, VLAN		Network Switching Technologies	4

Course Outcomes

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Apply basic concepts of IP Routing, Routing Protocols and Inter-VLAN Routing for a simple network.	15	Applying
CO2	Identify how to configure EIGRP and OSPF routing protocol for a simple network	14	Applying
CO3	Identify how to configure Multi-Area OSPF and IPv6 Routing for a simple network	15	Applying
CO4	Identify the concepts of Wide Area Network and the Evolution of Intelligent Network.	14	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Apply basic concepts of IP Routing, Routing Protocols and Inter-VLAN Routing for a simple network.		
M1.01	Outline the IP Routing Process	2	Understanding
M1.02	Identify how to configure Static Routing and Default Routing	3	Applying
M1.03	Summarize Dynamic Routing Protocols	2	Understanding
M1.04	Identify how to configure RIP	3	Applying
M1.05	Outline how routing between VLANs happens	2	Understanding
M1.06	Identify how to Configure Inter-VLAN Routing	3	Applying
Routing – IP Routing Process – Configuring IP Routing – Static Routing – Default Routing – Dynamic Routing – Administrative Distance – Routing Protocols – Routing Information Protocol (RIP) – Configuring RIP Routing Routing Between VLANs - Configuring Inter-VLAN Routing			
CO2	Identify how to configure EIGRP and OSPF routing protocol for a simple network		
M2.01	Summarize EIGRP Features and Operations	3	Understanding
M2.02	Identify how to configure EIGRP	3	Applying
M2.03	Outline OSPF Routing Protocol	2	Understanding
M2.04	Identify how to configure OSPF	3	Applying
M2.05	Identify how to configure Loopback Interfaces	3	Applying
	Series Test – I	1	

Contents: Enhanced IGRP – EIGRP Features and Operations – Neighbour Discovery – Reliable Transport Protocol – Diffusing Update Algorithm – Route Discovery and Maintenance – Configuring EIGRP – VLSM support and summarization – Controlling EIGRP Traffic – Split Horizon Open Shortest Path First – OSPF Basics – Terminology – Operation – LSA Flooding – SPF Tree calculation – OSPF Metrics – Configuring OSPF – Enabling – Configuring Area – Loopback Interfaces – Configuring Loopback Interfaces			
CO3	Identify how to configure Multi-Area OSPF and IPv6 Routing for a simple network		
M3.01	Outline categories of Multi-Area components	1	Understanding
M3.02	Identify how to configure basic multi-area OSPF	3	Applying
M3.03	Outline needs and features of IPv6	1	Understanding
M3.04	Summarize Hexadecimal number system, IPv6 notation and prefixes	1	Understanding
M3.05	Illustrate IPv6 Subnets and addressing types	2	Understanding
M3.06	Outline the working of IPv6 in an internetwork	2	Understanding
M3.07	Summarize IPv6 routing protocols	2	Understanding
M3.08	Identify how to configure IPv6 Routing for a simple network	3	Applying
Contents: Multi-Area OSPF – OSPF Scalability – Categories of Multi-Area Components – Adjacency Requirements – OSPF Router Roles – Link-State Advertisements – Hello Protocol – Neighbour States – Basic Multi-Area Configuration IPv6 – Need – Features – Hexadecimal Number System – IPv6 Addressing Notation – IPv6 Prefixes – IPv6 Subnets – IPv6 Addressing Types – Unicast – Multicast – Anycast – Working of IPv6 in an internetwork – Manual Address assignment – Stateless Autoconfiguration – DHCPv6 – IPv6 Routing Protocols – Static Routing with IPv6 – Configuring IPv6 Routing			
CO4	Identify the concepts of Wide Area Network and the Evolution of Intelligent Network.		
M4.01	Summarize Wide Area Networks	1	Understanding
M4.02	Outline Cabling the Serial Wide Area Network	1	Understanding
M4.03	Illustrate Virtual Private Networks and GRE Tunnels	2	Understanding
M4.04	Identify how to configure single-homed EBGp	2	Applying
M4.05	Outline Switch Stacking	1	Understanding
M4.06	Outline Cloud Computing and its effect on the Enterprise Network Service Models	2	Understanding
M4.07	Outline the Network Programmability in Enterprise Network	2	Understanding
M4.08	Summarize Application Program Interfaces	1	Understanding
M4.09	Outline APIC-EM and Intelligent WAN	2	Understanding

	Series Test – II	1	
Contents: Wide Area Networks – WAN Topology Options – Defining WAN Terms – WAN Connection Bandwidth – WAN Connection Types – Cabling the Serial WAN – Serial Transmission – DTE and DCE – Virtual Private Networks – Benefits of VPN – Enterprise and Provider Managed VPNs – IPsec – GRE Tunnels – Single-Homed EBGp – Configuring EBGp Evolution of Intelligent Networks – Switch Stacking – Cloud Computing and its Effect on the Enterprise Network – Cloud Service Models – Network Programmability – Application Programming Interfaces – APIC-EM – Intelligent WAN			

Text / Reference:

T/R	Book Title / Author
T1	Todd Lammle, <i>CCNA Routing and Switching Complete Study Guide</i> , 2 nd Edition, Sybex (Wiley) Publication, 2016
R1	Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. <i>Elements of Computer Networking: An Integrated Approach</i> . CareerMonk Publications, 2014.

Online Resources:

Sl.No	Website Link
1	https://www.freeccnastudyguide.com/study-guides/icnd2-200-101/
2	https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-tutorials/
3	https://www.guru99.com/introduction-ccna.html
4	https://study-ccna.com/
5	http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php